

Onat Gungor, Ph.D.

Assistant Professor

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Google Scholar

Research Profile

Trustworthy artificial intelligence for cyber-physical and autonomous systems, with emphasis on **efficient, secure, and reliable AI** in safety-critical environments. Research interests include edge AI, adversarial robustness, multimodal learning, LLM-based agents, cybersecurity, collaborative and autonomous systems, and embodied intelligence.

Academic Appointments

West Virginia University, Department of Computer Science and Electrical Engineering Assistant Professor; Director, Trustworthy, Efficient, and Reliable AI Systems (TERA) Lab	2026–Present
University of California San Diego, Computer Science and Engineering Postdoctoral Scholar, advised by Prof. Tajana Rosing	2024–2026
UC San Diego & San Diego State University, Electrical and Computer Engineering Graduate Student Researcher, advised by Prof. Tajana Rosing and Prof. Baris Aksanli.	2019–2023

Education

Ph.D., Electrical and Computer Engineering University of California San Diego & San Diego State University Joint Doctoral Program. Dissertation: <i>Towards Intelligent, Secure, and Efficient Industrial Internet of Things.</i>	2023
B.S., Computer Science and Engineering Ozyegin University. High Honor Degree; graduated at the top of the department.	2019
B.S., Industrial Engineering Ozyegin University. Valedictorian Award.	2018

Peer-Reviewed Publications

Journal Articles

1. J. Chen, **O. Gungor**, Z. Shang, and T. Rosing, “SAGE: Sample-Aware Guarding Engine for Robust Intrusion Detection Against Adversarial Attacks,” *Computer Networks*, 2026.
2. C. I. Kocal, **O. Gungor**, T. Rosing, and B. Aksanli, “ReLATE+: Unified Framework for Adversarial Attack Detection, Classification, and Resilient Model Selection in Time-Series Classification,” *Electronics*, 2026.
3. S. Nayak, **O. Gungor**, and T. Rosing, “Security in Collaborative Driving: A Survey of Threats, Defenses, and Emerging Trends,” *Electronics*, 2026.
4. E. Li, **O. Gungor**, Z. Shang, J. Chen, and T. Rosing, “CITADEL: Continual Anomaly Detection for Enhanced Learning in IoT Intrusion Detection,” *Computer Networks*, 2026.
5. **O. Gungor**, T. Rosing, and B. Aksanli, “STEWART: Stacking Ensemble for White-Box Adversarial Attacks Towards More Resilient Data-Driven Predictive Maintenance,” *Computers in Industry*, 2022.
6. **O. Gungor**, T. Rosing, and B. Aksanli, “DOWELL: Diversity-Induced Optimally Weighted Ensemble Learner for Predictive Maintenance of Industrial Internet of Things Devices,” *IEEE Internet of Things Journal*, 2021.
7. **O. Gungor**, T. Rosing, and B. Aksanli, “RESPIRE++: Robust Indoor Sensor Placement Optimization under Distance Uncertainty,” *IEEE Sensors Journal*, 2021.
8. **O. Gungor**, B. Aksanli, and R. Aydogan, “Algorithm Selection and Combining Multiple Learners for Residential Energy Prediction,” *Future Generation Computer Systems*, 2019.

Conference Proceedings

1. N. P. Pandey, J. Park, J. Kong, **O. Gungor**, and T. Rosing, “NgramRL: Accelerating RL Post-Training for Structured Generation via Rollout-Aware N-gram Speculative Decoding,” *IEEE/ACM ASP-DAC*, 2027.
2. S. Sharma, N. P. Pandey, **O. Gungor**, and T. Rosing, “LIMBO: Lifelong Inference-Time Memory and Budget Optimization for LLM Agents,” *IEEE ICTAI*, 2026.
3. I. Delgado, H. Gupta, L. Shamoon, B. Khatri, N. Sapre, **O. Gungor**, and T. Rosing, “T-SMART: Mechanism-Level Attribution for Tool-Augmented Time-Series Question Answering,” *IEEE ICTAI*, 2026.
4. Y. Tian, Z. Wang, **O. Gungor**, X. Fan, and T. Rosing, “LifeAgentBench: A Multi-dimensional Benchmark and Agent for

- Personal Health Assistants in Digital Health,” *EMNLP*, 2026.
5. B. Gupta, **O. Gungor**, and T. Rosing, “G-MARK: Grounded Multi-Agent Reasoning for Cooperative Driving via Knowledge Graphs,” *IEEE ICMLA*, 2026.
 6. **O. Gungor**, R. Sood, J. Zhou, and T. Rosing, “EAGER: Edge-Aligned LLM Defense for Robust, Efficient, and Accurate Cybersecurity Question Answering,” *IEEE ISNCC*, 2026.
 7. W. Lee, P. Mekkoth, Y. Tian, **O. Gungor**, and T. Rosing, “FoodCHA: Multi-Modal LLM Agent for Fine-Grained Food Analysis,” *IEEE ISNCC*, 2026.
 8. G. K. Tokgoz, **O. Gungor**, T. Rosing, and B. Aksanli, “INTARG: Informed Real-Time Adversarial Attack Generation for Time-Series Regression,” *IEEE ISNCC*, 2026.
 9. N. P. Pandey, J. Park, **O. Gungor**, F. Ponzina, and T. Rosing, “QMC: Efficient SLM Edge Inference via Outlier-Aware Quantization and Emergent Memories Co-Design,” *IEEE/ACM ISLPED*, 2026.
 10. J. Chen, A. Deevi, **O. Gungor**, and T. Rosing, “CAN-QA: A Question-Answering Benchmark for Reasoning over In-Vehicle CAN Traffic,” *IEEE ICCCN*, 2026.
 11. **O. Gungor**, I. Kale, J. Zhou, and T. Rosing, “LIGHT-HIDS: A Lightweight and Effective Machine Learning-Based Framework for Robust Host Intrusion Detection,” *IEEE ICMLA*, 2025.
 12. **O. Gungor**, R. Sood, H. Wang, and T. Rosing, “AQUA-LLM: Evaluating Accuracy, Quantization, and Adversarial Robustness Trade-offs in LLMs for Cybersecurity Question Answering,” *IEEE ICMLA*, 2025.
 13. Y. Tian, **O. Gungor**, X. Yu, and T. Rosing, “DailyLLM: Context-Aware Activity Log Generation Using Multi-Modal Sensors and LLMs,” *IEEE MASS*, 2025.
 14. N. P. Pandey, S. Kulkarni, D. Wang, **O. Gungor**, F. Ponzina, and T. Rosing, “DPQ-HD: Post-Training Compression for Ultra-Low Power Hyperdimensional Computing,” *ACM GLSVLSI*, 2025.
 15. C. I. Kocal, **O. Gungor**, A. Tartz, T. Rosing, and B. Aksanli, “ReLATE: Resilient Learner Selection for Multivariate Time-Series Classification Against Adversarial Attacks,” *IEEE CSR*, 2025.
 16. S. Fuhrman, **O. Gungor**, and T. Rosing, “CND-IDS: Continual Novelty Detection for Intrusion Detection Systems,” *ACM/IEEE DAC*, 2025.
 17. L. Zhang, Q. Zhao, R. Wang, S. Bian, **O. Gungor**, F. Ponzina, and T. Rosing, “ORCA: Offload Rethinking by Cloud Assistance for Efficient Environmental Sound Recognition on LPWANs,” *ACM SenSys*, 2025.
 18. L. Zhang, F. Ponzina, **O. Gungor**, and T. Rosing, “E-QUARTIC: Energy Efficient Edge Ensemble of Convolutional Neural Networks for Resource-Optimized Learning,” *IEEE/ACM ASP-DAC*, 2025.
 19. **O. Gungor**, A. Rios, P. Mudgal, N. Ahuja, and T. Rosing, “A Robust Framework for Evaluation of Unsupervised Time-Series Anomaly Detection,” *ICPR*, 2024.
 20. F. Asgarinejad, F. Ponzina, **O. Gungor**, T. Rosing, and B. Aksanli, “HDXpose: Harnessing Hyperdimensional Computing’s Explainability for Adversarial Attacks,” *ACM/IEEE ICCAD*, 2024.
 21. **O. Gungor**, T. Rosing, and B. Aksanli, “A2HD: Adaptive Adversarial Training for Hyperdimensional Computing-Based Intrusion Detection Against Adversarial Attacks,” *IEEE CSR*, 2024.
 22. **O. Gungor**, E. Li, Z. Shang, Y. Guo, J. Chen, J. Davis, and T. Rosing, “Rigorous Evaluation of Machine Learning-Based Intrusion Detection Against Adversarial Attacks,” *IEEE CSR*, 2024.
 23. **O. Gungor**, T. Rosing, and B. Aksanli, “ROLDEF: Robust Layered Defense for Intrusion Detection Against Adversarial Attacks,” *DATE*, 2024.
 24. X. Yu, M. Zhou, F. Asgarinejad, **O. Gungor**, B. Aksanli, and T. Rosing, “Private and Secure Edge AI with Hyperdimensional Computing,” *ACM/IEEE DAC*, 2023.
 25. M. Timken, **O. Gungor**, T. Rosing, and B. Aksanli, “Analysis of Machine Learning Algorithms for Cyber Attack Detection in SCADA Power Systems,” *IEEE SmartNets*, 2023.
 26. **O. Gungor**, T. Rosing, and B. Aksanli, “HD-I-IoT: Hyperdimensional Computing for Resilient Industrial Internet of Things Analytics,” *DATE*, 2023.
 27. **O. Gungor**, T. Rosing, and B. Aksanli, “DENSE-DEFENSE: Diversity Promoting Ensemble Adversarial Training Towards Effective Defense,” *IEEE SENSORS*, 2022.
 28. **O. Gungor**, T. Rosing, and B. Aksanli, “CAHEROS: Constraint-Aware Heuristic Approach for Robust Sensor Placement,” *IEEE SENSORS*, 2021.
 29. **O. Gungor**, T. Rosing, and B. Aksanli, “ENFES: Ensemble Few-Shot Learning for Intelligent Fault Diagnosis with Limited Data,” *IEEE SENSORS*, 2021.
 30. **O. Gungor**, T. Rosing, and B. Aksanli, “OPELRUL: Optimally Weighted Ensemble Learner for Remaining Useful Life Prediction,” *IEEE ICPHM*, 2021.
 31. **O. Gungor**, J. Garnier, T. Rosing, and B. Aksanli, “LENARD: Lightweight Ensemble Learner for Medium-Term Electricity Consumption Prediction,” *IEEE SmartGridComm*, 2020.
 32. **O. Gungor**, T. Rosing, and B. Aksanli, “RESPIRE: Robust Sensor Placement Optimization in Probabilistic Environments,” *IEEE SENSORS*, 2020. **Best Paper Nominee**.

Workshop Papers and Extended Abstracts

1. J. Chen, **O. Gungor**, Z. Shang, E. Li, and T. Rosing, “DYNAMITE: Dynamic Defense Selection for Enhancing Machine Learning-Based Intrusion Detection Against Adversarial Attacks,” *IEEE/ACM Workshop on the Internet of Safe Things*, 2025. **Best Paper Award**.
2. **O. Gungor**, A. Rios, N. Ahuja, and T. Rosing, “TS-OOD: An Evaluation Framework for Time-Series Out-of-Distribution

- Detection and Prospective Directions for Progress,” *AAAI Workshop on AI for Time Series Analysis*, 2025.
3. E. Li, Z. Shang, **O. Gungor**, and T. Rosing, “SAFE: Self-Supervised Anomaly Detection Framework for Intrusion Detection,” *AAAI Workshop on AI for Cyber Security*, 2025.
 4. **O. Gungor**, T. Rosing, and B. Aksanli, “Adversarial-HD: Hyperdimensional Computing Adversarial Attack Design for Secure Industrial Internet of Things,” *IEEE/ACM Workshop on the Internet of Safe Things*, 2023. **Best Paper Runner-Up**.
 5. **O. Gungor**, U. Cakan, R. Aydogan, and P. Ozturk, “Effect of Awareness of Other Side’s Gain on Negotiation Outcome, Emotion, Argument and Bidding Behavior,” *International Workshop on Agent-Based Complex Automated Negotiation*, 2019. **Best Student Paper**.
 6. Y. Tian, **O. Gungor**, X. Yu, and T. Rosing, “Fine-Grained Contextualized Activity Logs Generation Based on Multi-Modal Sensor Data and LLM,” Poster Abstract, *ACM SenSys*, 2025.

Honors and Awards

- **Best Paper Award**, IEEE/ACM Workshop on the Internet of Safe Things, 2025 (DYNAMITE).
- **Best Paper Runner-Up**, IEEE/ACM Workshop on the Internet of Safe Things, 2023 (Adversarial-HD).
- San Diego State University **University Graduate Fellowship**, 2021.
- **Best Paper Nominee**, IEEE SENSORS, 2020 (RESPIRE).
- **Best Student Paper**, International Workshop on Agent-Based Complex Automated Negotiation (ACAN), 2019.

Selected Research Collaborations

- **Intel Labs**: time-series anomaly detection, phase detection, and out-of-distribution detection.
- **Google**: multimodal and LLM-powered personal health agent systems.
- **Lawrence Berkeley National Laboratory**: automated large-scale network traffic classification.

Teaching Experience

- **Instructor, West Virginia University**: CPE 442 – Introduction to Digital Computer Architecture (Fall 2026).
- **Instructor, UC San Diego**: CSE 15L – Software Tools and Techniques (Spring 2022, Fall 2023); CSE 140 – Components and Design Techniques for Digital Systems (Summer 2022).
- **Teaching Assistant**: SDSU COMPE 375 – Embedded Systems Programming (Summer 2021, Summer 2022); UCSD CSE 140 – Components and Design Techniques for Digital Systems (Winter 2022).

Student Mentoring and Advising

Mentored **40+ students since 2020** across Ph.D., M.S., and undergraduate research. Selected mentees include:

- Nilesh Pandey (Ph.D.) – efficient and scalable edge AI; publications at GLSVLSI 2025, ISLPED 2026, ASP-DAC 2027.
- Ye Tian (Ph.D.) – LLMs and physical-world understanding; publications at SenSys 2025, MASS 2025, EMNLP 2026.
- Sean Fuhrman (M.S.) – continual novelty detection for network intrusion detection; DAC 2025.
- Le Zhang (M.S.) – efficient edge AI and edge-cloud collaboration; ASP-DAC 2025 and SenSys 2025.
- Ipek Kocal (M.S.) – resilient time-series classification under adversarial attacks; CSR 2025 and Electronics 2026.
- Jing Chen, Elvin Li, Zhengli Shang, Ishaan Kale, Jiasheng Zhou, Roshan Sood, and Harold Wang (undergraduate researchers) – multiple peer-reviewed publications spanning adversarial ML, intrusion detection, and LLM security.

Selected Invited Talks

- “Collaborative Intelligence for Securing the IoT: Robust, Adaptive, and Secure AI-Driven Intrusion Detection Systems,” SRC JUMP 2.0 CoCoSys Theme 4 Meeting, May 2025.
- “AI-Driven Cybersecurity: Advancing Intrusion Detection, LLM Security and Time-Series Anomaly Detection,” Cisco Foundation AI Group, April 2025.

Academic Service and Outreach

Journal Reviewer: IEEE Internet of Things Journal; IEEE Transactions on Information Forensics and Security; IEEE Transactions on Dependable and Secure Computing; IEEE Transactions on Industrial Informatics; IEEE Transactions on Industrial Electronics; IEEE Transactions on Network and Service Management.

Conference Reviewer: DATE 2027; ASPLOS 2025; DATE 2025; ECAI 2024–2025; MICRO 2024; CODES+ISSS 2023; DAC 2023; IEEE SENSORS 2022.

Research Outreach: ENLACE bi-national summer research program; UC San Diego Early Research Scholars Program.